

Problem I. Olympiad for Everyone

Time Limit 3000 ms

Mem Limit 262144 kB

OS Windows

It was a rainy autumn day. A total of n students attended an olympiad, their intelligence can be ranked with integers a_i . And what a coincidence! Jury prepared exactly n problems, their difficulty can be ranked with integers b_i .

A student can solve a problem if and only if their intelligence a_i is greater or equal than the difficulty b_j of the problem. Jury don't want to upset participants, so each participant was given one problem they can solve.

Count the amount of ways to distribute problems between students, so that everyone is able to solve the problem they got. Each student should get exactly one problem, and each problem should be given to exactly one student. The answer can be large, so divide the answer by p and output the remainder (compute the answer modulo p).

Input

The first line contains a single integer n — the amount of participants ($1 \leq n \leq 10^5$).

The second line contains n space-separated integers a_i — values of intelligence of participants ($0 \leq a_i \leq 10^9$).

The third line contains n space-separated integers b_i — values of difficulty of problems ($0 \leq b_i \leq 10^9$).

The fourth line contains one integer p ($1 \leq p \leq 10^9$).

Output

Output a single integer — the amount of ways to distribute the problems according to the statement modulo p .

Sample

Input	Output
3 1 2 3 1 2 3 100	1

Input	Output
3 3 3 3 1 1 1 100	6